

Power System Simulation with LTspice

A Practical Guide to Modeling and Simulation with LTspice

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Version 1.0

Power System Simulation with LTspice
Practical Examples, Simulation and Models

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For everyone who wants to deeper understand power systems

Preface

This book presents practical methods for modelling and analysing electrical power systems with LTspice. The emphasis is placed on reproducible examples that connect physical interpretation, manual calculation and numerical simulation.

The material is intended for students, engineers and technically interested readers. Each chapter combines a concise theoretical introduction with worked calculations, circuit models, simulation results and practical remarks.

The book is divided into 9 simulation examples, ranging from simple to more complex. The formatting of the book and, in some cases, the translation from German to English were done with the help of ChatGPT. All examples and technical details are, of course, provided by the author. I wish readers every success with their simulations and hope that they, too—like me—will find great enjoyment in them. I would also like to thank the developers of LTspice for such a wonderful piece of software that made it possible for me to write this book. Although it is designed for simulating electronic circuits, it is so flexible that it allowed me to perform high-voltage analyses with it as well. Using a mathematical trick, I was also able to adapt LTspice for steady-state load-flow calculations, which led to the development of my second toolbox. This toolbox is also freely available and will be presented in more detail in a future book.

Friedrich Schmidt
Austria, July 2026

Acknowledgements

I would like to express my sincere thanks for the wonderful examples in the book "Dynamic Behavior of Electrical Machines" by Prof. Otto Justus, which provided me with a great deal of inspiration for my own book.

How to Use This Book

The chapters are written so that the reader can reproduce the examples directly. The following elements are used consistently:

Definition	Introduces an important technical term.
Remark	Adds interpretation, limitations or practical advice.
Example	Presents a worked application.
Calculation	Aligns formulas, substituted values and results.
Simulation	Describes model settings and simulation steps.
Chapter Summary	Collects the main conclusions at the end of a chapter.

Project files are stored in the folders `ltspice/` and `figures/`

It is recommended that you work through the book from beginning to end, always in parallel with simulations using the LTspice simulator. A prerequisite for successful simulations is a good understanding of how to use the LTspice simulator. A solid understanding of Spice syntax is absolutely essential for a deep understanding of the constructions of the models. The complex models of the transformer and the synchronous generator are explained in the appendix of the book. To better understand the detailed modeling, it is recommended to review the LTspice .lib library files.

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About this book

A practical introduction for everyone interested in the simulation of high-voltage electrical power networks.

Over the past 25 years, I have developed two educational toolboxes for LTspice. The first supports nonlinear load-flow calculations; the second is designed for the transient simulation of electrical power networks.

The toolboxes contain the essential components required to build realistic power-system models:

- buses and sources;
- synchronous machines and transformers;
- overhead lines and power cables;
- loads and shunt reactors; and
- surge-protection devices, Petersen coils, and many other elements.

Some elements provide parameters that can be changed directly by the user. Others are supplied with carefully selected presets, which can also be overwritten whenever a study requires different values.

The transformer library ranges from simple models to sophisticated representations with a nonlinear iron core, hysteresis, and remanent flux. Eight vector groups and voltage levels from 400 kV down to 400 V are included. Additional vector groups can be created by editing the macro-models, because all `.lib` files are freely accessible and editable.

Hundreds of examples for both toolboxes are available on my website, ranging from simple introductory circuits to complex networks. Many schematic files contain their own documentation, and the first nine examples are explained in detail in this book.

These toolboxes are especially dedicated to students of electrical engineering, but they are equally useful to teachers, engineers, and anyone who wants to explore power-system behaviour through simulation.

I hope you enjoy experimenting with these simulations as much as I do. Your feedback is always welcome.